UMC Data Hackathon:
"Bridging the Digital Divide by Uncovering Digital Deserts"

October 2025 - September 2026

Global Gateway
Funded by the European Union

UMC Data Hackathon

Overview

• Designed to mobilize creativity & data analysis to identify connectivity gaps and populations left behind in achieving **Universal & Meaningful Connectivity (UMC).**

Through **data processing, visualization and storytelling**, teams will:

- Identify areas with limited connectivity
- Profile populations excluded from meaningful connectivity
- Explore innovative statistical methods with diverse datasets
- Help design policies to achieve **Universal and Meaningful Connectivity**

• Organized by **ITU** (BDT/IDA, Regional Office Asia-Pacific, Area Office for South Asia & Innovation Centre New Delhi) with financial support from the **European Union**.

• Side event of the **World Telecommunication/ICT Indicators Symposium (WTIS) 2026**, in Geneva, Switzerland.

Global Gateway
Funded by the European Union
ITU

UMC Data Hackathon

Key Information

Who will participate

- Worldwide students & young professionals (≤30 years)
- **104 teams** of 3–5 (multidisciplinary, gender-diverse: 35% women)

Methods

- **Mentorship** & jury by ITU staff and experts
- **Data** provided on infrastructure, coverage, user-generated, socio-economic & geospatial datasets
- Other data can be found and used by teams
- Participants use their own laptops & preferred software

Deliverables

- Maps, dashboards, policy recommendations, open code

Recognition & Awards

- Certificates of participation
- Publication of results on social media and WTIS-26 report
- Winner team invited to present at WTIS-26 (travel and subsistance costs covered)

Global Gateway
Funded by the European Union

UMC Data Hackathon

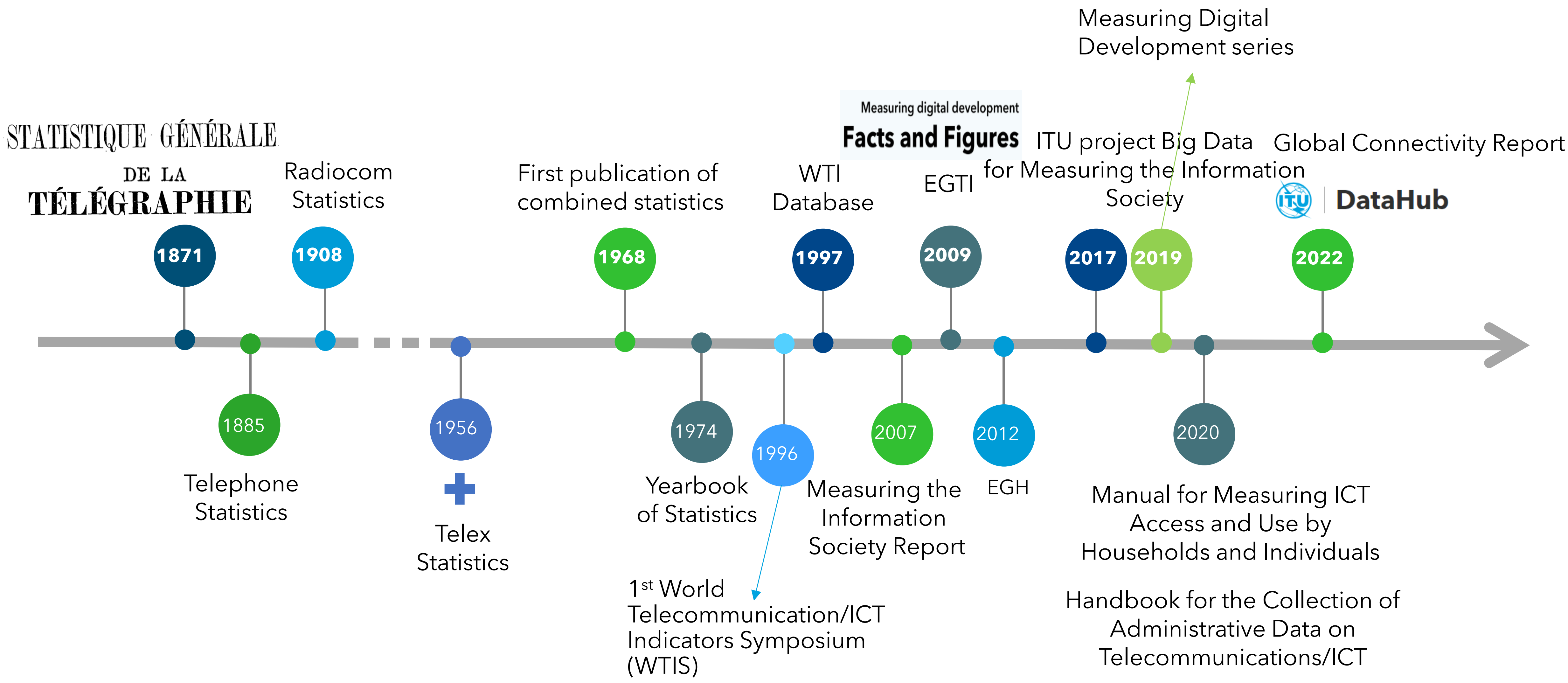
Timeline

Date	Action
September 2025	Announcement
October–December 2025	Circular to Member States and Academia Members + Communication + Registration of teams **CLOSED 31 December 2025** **104 teams successfully registered!!**
January 2025–March 2026	Webinar 1: introduction to the concept of Universal and Meaningful Connectivity and its statistical measurement ; examples of digital policies **TODAY** Webinar 2: explanation of data sets; Statistical methodology **9-16 February 2026 (TBC)**
End April 2026	Two-day physical meeting session (**New Delhi**) for shortlisted teams and submission of Projects **28-29 April 2026**
May 2026	Evaluation of Projects
September 2026	Winners present at WTIS 2026 (**Geneva**) **September 2026**

Global Gateway
Funded by the European Union
ITU

What does the ICT do in the field of statistics?

History of ICT statistics at ITU



Global Gateway
Funded by the European Union

Leading the global ICT statistics agenda

The International Telecommunication Union (ITU) leads the global ICT statistics agenda, supporting evidence-based decision-making to drive universal and meaningful connectivity and sustained digital transformation. ITU's statistical activities cover the entire data lifecycle. The ICT Data and Analytics Division (IDA), part of ITU-D's Digital Knowledge Society Department, carries out most of this work.

Statistical standards

Through the Expert Group on Telecom/ICT Indicators (EGTI) and the Expert Group on ICT Household Indicators (EGH), ITU defines and updates international statistical standards for ICT indicators.

more

Data collection & dissemination

IDA compiles statistics for hundreds of ICT indicators, based on data collected from over 200 economies; it computes global, regional and country group estimates. All data is available for free on the ITU DataHub.

more

Analysis & research

Publications of the *Measuring Digital Development* series assess the state of global connectivity, tackle specific themes, and identify solutions.

more

Data science for official statistics

IDA's data science practice leverages big data and new methods to enhance the accuracy, timeliness, and granularity of ICT statistics.

more

Capacity development & technical assistance

IDA supports the statistical community and other stakeholders, including policymakers, by developing technical documentation, training materials, online courses, workshops, and providing technical assistance.

more

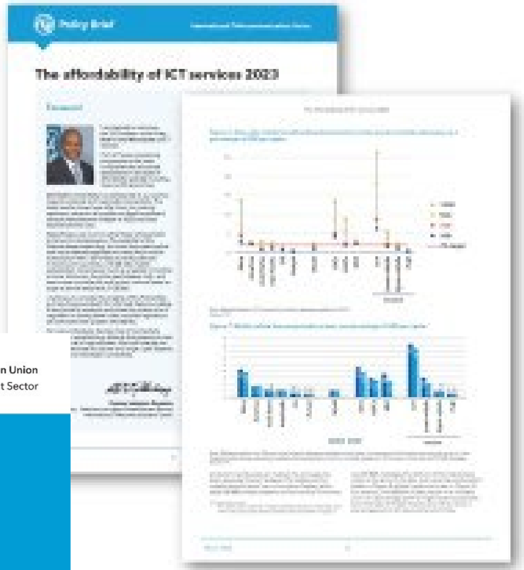
Global cooperation & engagement

IDA organises events, including WTIS, the premier global event on ICT statistics, and cooperates with various organizations, to mobilize resources, advance the statistics agenda, leverage synergies, scale up initiatives, and maximize impact.

more

Global Gateway
Funded by the European Union

Data analysis and research reports



Global Gateway

Funded by
the European Union

What does Universal and Meaningful Connectivity mean?

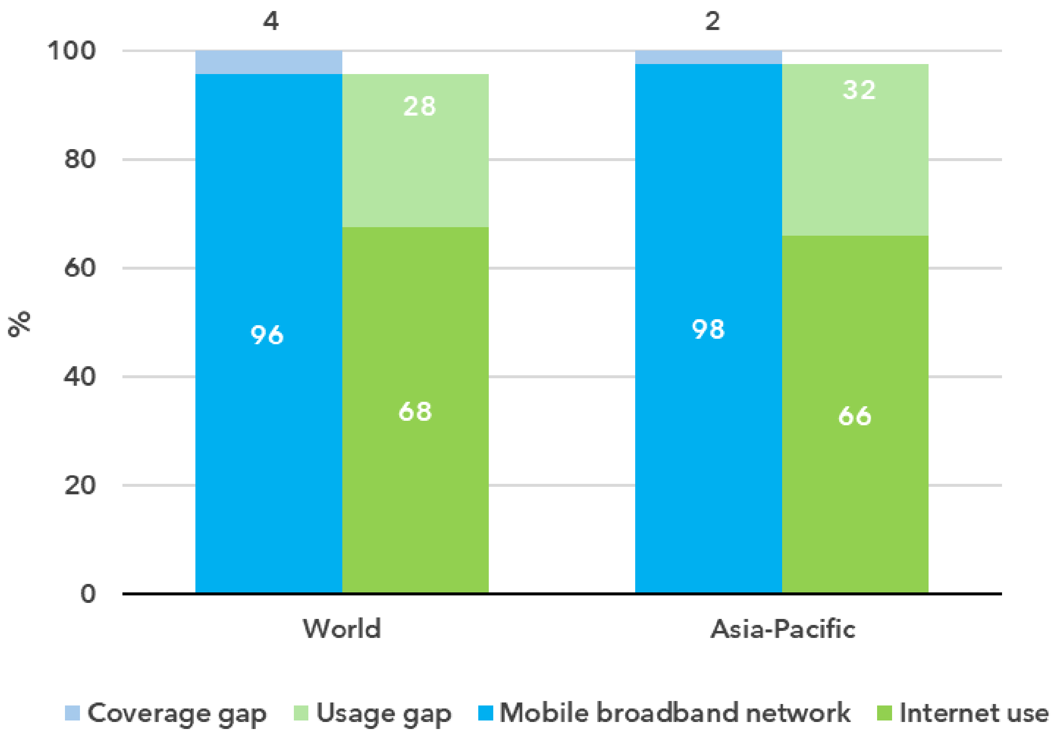
Coverage gap is much lower than usage gap

Coverage gap = people without access to a fixed or mobile broadband network

Usage gap = People that do not use the Internet while having coverage

Why don't people use the Internet?

When they use it, is it meaningful?



Global Gateway

Funded by the European Union

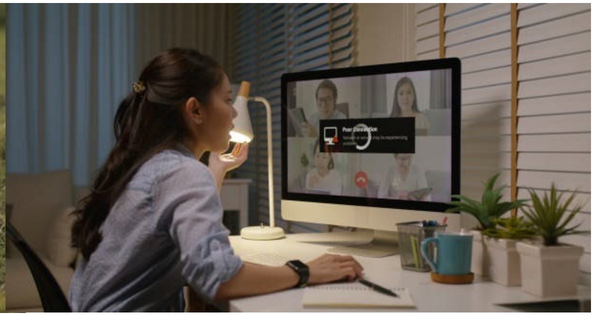
ITU

Universal and Meaningful Connectivity #UMC

“The possibility for everyone to enjoy a safe, satisfying, enriching, productive, and affordable online experience.”

Global Gateway
Funded by the European Union

What happens if there is no **#UMC**?



Increased cost of digital exclusion

Barriers to economic development

Increased vulnerability to global challenges

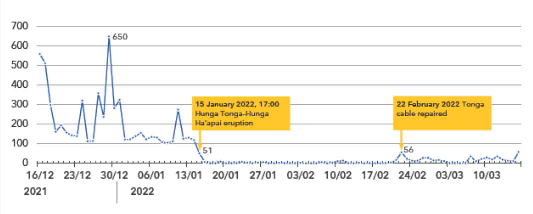
Limited access to essential services

Deepening social and economic inequalities

Global Gateway
Funded by the European Union

Figure 10.2 Disruptions in Tonga's telecommunication networks

Data points per day from Speedchecker, Tonga, 16 December 2021–16 March 2022



Source: Speedchecker.

Figure 10.3 DCM map for Tonga before and after the volcanic eruption and tsunami

Before, on 13 January 2022

After, on 7 February 2022

Note: White squares show baseline connectivity, all measurement datapoints received since the start of the measurement campaign on 16 December 2021. Red squares show the connectivity datapoints seen during the last 24-hour period.
Source: ITU.

Crises disrupt #UMC …

Global Gateway

Funded by the European Union

ITU

... but #UMC helps in crises

coordinate rescue efforts

disseminate critical information to affected populations

allow **survivors to communicate** with their loved ones or request assistance through voice calls, social media and messaging apps

provide **access to education, online language** learning ration into new communities

enable access to **virtual mental health support online legal resources** for affected individual

allow for remote **healthcare consultations**, **contact tracing** via mobile applications

enable **weather forecasts** and the **monitoring of agricultural conditions**

facilitate **online marketplaces** for food distribution,

allows for **fundraising** efforts to support affected communities

History of the concept of UMC

2018-2019: UN SG convenes a **High-level Panel on Digital Cooperation** which drafts the report "**The age of digital interdependence**".

2020: UN SG issues his report **Roadmap for Digital Cooperation**, which includes, at its core, a **commitment to "connect" all people to the Internet.**

2021: Roundtable on Global Connectivity, co-chaired by **UNICEF** and **ITU**, with the support of the Office of the Secretary-General's Envoy on Technology (**OSET**), to **follow up on the Roadmap**

2022: OSET and ITU announced a new set of **UN targets for universal and meaningful digital connectivity** to be achieved by 2030.

2024: **UN General Assembly** adopts the Pact for the Future and **Global Digital Compact**

Global Gateway

Funded by the European Union

How do we measure progress towards Universal and Meaningful Connectivity?

#UMC measurement questions

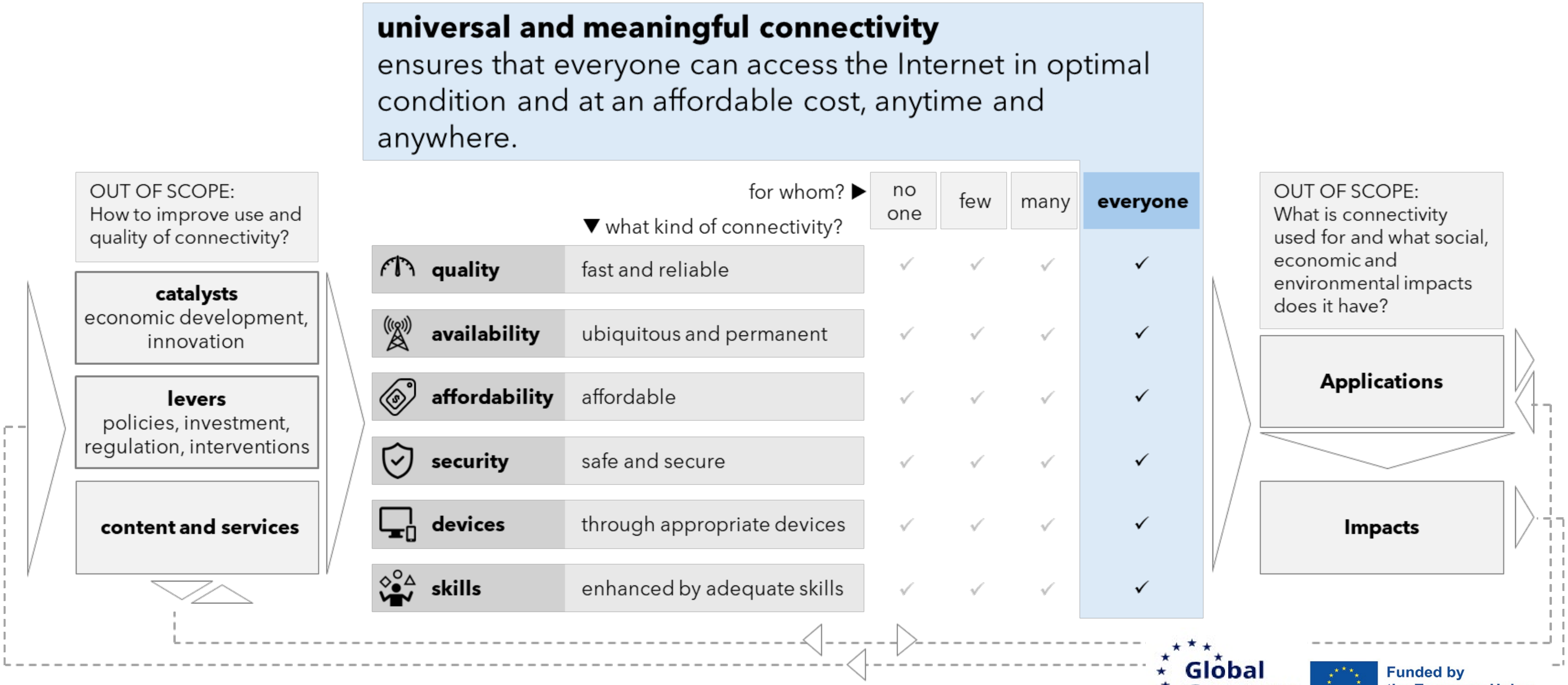
Dimensions	Conceptual questions	Measurement objectives
Connection quality	Do people have access to **high-speed, stable Internet connections** suitable for their specific needs and activities online?	Assessing the speed, reliability, and stability of Internet connections.
Availability for use	Are people able to use the Internet as **frequently** and intensively as they wish? Can people access the Internet in **different locations**, wherever and whenever they want?	Measuring the regularity and intensity of Internet use among individuals. Evaluating the accessibility and convenience of Internet use in various contexts and locations.
Affordability	Are **Internet access, devices, and data plans affordable** and sufficient relative to people's incomes, allowing for flexible and desired quality of use?	Evaluating the affordability, adequacy, and flexibility of Internet services relative to individual incomes.
Devices	Do people have access to the **appropriate devices** necessary to fully engage with and benefit from digital opportunities?	Evaluating the availability, variety, and suitability of devices used to access the Internet.
Digital skills	Do people possess the necessary **skills** to leverage digital opportunities and manage potential risks effectively?	Assessing individuals' competency and confidence in using the Internet effectively.
Safety and security	Do people have access to **secure Internet connections**, can they **navigate online safely**, and do they **feel secure** in their online interactions and activities?	Assessing the safety and security of user online experience including concerns and exposure to harmful content and to-enabled crime.

Global Gateway
Funded by the European Union

The dimensions of UMC

how the different dimensions of UMC are integrated and how are statistics used for policy design, monitoring and evaluation?

Framework for universal and meaningful connectivity



Global Gateway

Funded by the European Union

Barriers to #UMC : 5 divides

- *Geography* : ***urban / rural gap***
  - Infrastructure gap: Fibre-optic cables, connection to submarine cables, Satellite coverage
  - Population density - Remote areas, mountains, preventing overland cables
- *Demography*
  - Older populations **age gap**
  - **Gender gap**
- ***Income gap***
- ***Education gap*** *(closely related to demography, gender and location)*
- ***Cultural and behavioural barriers*** : *language of contents, behavioural aspects*
- *Some population groups are at the* ***intersect of the barriers***
  - Lower rural usage is partly a result of a lack of infrastructure, but there are additional factors at play: lower income levels, lower levels of education and of ICT skills
  - Refugees, IDPs, crisis-affected people
  - Homeless, marginalised and vulnerable groups (older, young, with disabilities, indigenous, etc.)

→ **Hence, the need for disaggregated indicators**

Global Gateway
Funded by the European Union

- *Define the **concept of UMC** in a plain and standardized "statisticians' language":*
  - Terms
  - Dimensions
  - Indicators
- ***Generate statistical information** and indicators to estimate the progress towards UMC*
  - Compiling what exists
  - Piloting new indicators
  - Setting standards with established practices
- *Create **capacity in producing and using statistical information about UMC***
  - Focus on developing and transition countries where statistical information is scarce
- *Strengthen regional **collaboration and coordination between producers and users** of the statistical information*

Global Gateway
Funded by the European Union

#UMC Targets

Country-level indicator	Threshold to meet the target	Target share of countries (for which data is available)
share of population aged 15+ who uses the Internet	> 95%	90%
gender parity score The gender parity score is the share of women using the Internet among the female population aged 15+ divided by the share of men using the Internet among the male population aged 15+	> 0.98	90%
share of households with Internet access	> 95%	90%
share of businesses (with 10+ staff) that use the Internet	=100%	90%
share of primary schools connected to the Internet	=100%	90%
share of secondary schools connected to the Internet	=100%	90%
share of population aged 15+ who owns a mobile phone	>95%	90%
price of the fixed-broadband Internet price basket	< 2% of GNI per capita	90%
price of the mobile-broadband Internet price basket	< 2% of GNI per capita	90%

Global Gateway
Funded by the European Union

What can be done to progress towards Universal and Meaningful Connectivity?

Reasons vs opinions

No single evidence is perfect: combine multiple inputs for a full picture:

- Expert knowledge
- Published research
- Stakeholder consultations
- Policy evaluations
- Costing information
- **Output from statistical operations and statistical modelling…**

Global Gateway
Funded by the European Union